

Resident Thermal Comfort & Air-Conditioning Energy - Operational Lessons

Harmony Senior Residences · Senior Living · Singapore

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1. Background

Residents at Harmony Senior Residences are aged 70 to 95. Ageing reduces the body's ability to regulate temperature, so residents are more sensitive both to cold draughts and to heat stress. Before this initiative, room thermostats could be set anywhere from 16°C to 30°C. Visitors and staff often set rooms to 20°C, residents then felt cold at night, and common areas felt stuffy during the day. Relative humidity above 70% in some rooms caused mould on walls and wardrobes.

2. Measures

- Comfort band: thermostats limited to 24-26°C; residents can adjust within the band, and nursing staff can approve medical exceptions.
- Humidity control: fan coil units use a dehumidification mode at night, keeping relative humidity below 65%.
- Ceiling fans in rooms and lounges provide air movement so that 25-26°C feels comfortable.
- Night setback: common-area setpoints rise by 1°C after 11 pm.

3. Engagement

- Residents and families were briefed at town halls before any change, and each resident's preferences were recorded in their care plan.
- Care staff were trained to recognise signs of heat stress and of feeling cold, and to report comfort issues through the maintenance app.
- A monthly comfort check was added to the care routine.

4. Results (six months)

KPI	Before	After
Air-conditioning energy (MWh/month)	168	148
Comfort complaints per month	25	15
Rooms with mould reports	9	2

5. Lessons learned

- Involve residents and families before changing anything they can feel; explain why and listen.
- Humidity matters as much as temperature.
- Health comes first: medical needs override energy targets, and exceptions must be easy to approve.
- Make small changes and measure comfort as carefully as energy.